

Passages - A Brief History of Time

One can now see why all bodies fall at the same rate: a body of twice the weight will have twice the force of gravity pulling it down, but it will also have twice the mass. According to Newton's second law, these two effects will exactly cancel each other, so the acceleration will be the same in all cases.

Maxwell's theory predicted that radio or light waves should travel at a certain fixed speed.

An event is something that happens at a particular point in space and at a particular time.

Given an event P, one can divide the other events in the universe into three classes. Those events that can be reached from the event P by a particle or wave traveling at or below the speed of light are said to be in the future of P. They will lie within or on the expanding sphere of light emitted from the event P. Thus they will lie within or on the future light cone of P in the space-time diagram. Only events in the future of P can be affected by what happens at P because nothing can travel faster than light.

Similarly, the past of P can be defined as the set of all events from which it is possible to reach the event P traveling at or below the speed of light. It is thus the set of events that can affect what happens at P. The events that do not lie in the future or past of P are said to lie in the elsewhere of

P. What happens at such events can neither affect nor be affected by what happens at P. For example, if the sun were

to cease to shine at this very moment, it would not affect things on earth at the present time because they would be in the elsewhere of the event when the sun went out.

When we look at the universe, we are seeing it it was in the past.

In 1915 Einstein proposed the general theory of relativity, he suggested that gravity is not a force like other forces but is a consequence of the fact that space time is curved or warped by the distribution of mass and energy in it.

In general relativity bodies always follow straight lines in four-dimensional space time, but they nevertheless appear to us to move along, curved paths in our three-dimensional space.

Another prediction of general relativity is that time should appear to run slower near a massive body like the earth. This is because there is a relation between the energy of light and its frequency (that is, the number of waves of light per second): the greater the energy, the higher the frequency. As light travels upward in the earth's gravitational field, it loses energy, and so its frequency goes down. (This means that the length of time between one wave crest and the next goes up.)

Most galaxies appear red-shifted (all moving away - Doppler Effect). The size of a galaxy's red shift is directly proportional to the galaxy's distant from us. Or, in other words, the farther a galaxy is, the fast it is moving away! Meaning the universe was not static, but is in fact expanding; the distance between the different galaxies growing all the time.

Paradoxically, the more fuel a star starts off with, the sooner it runs out. This is because the more massive the star is, the hotter it needs to be to balance its gravitational attraction. And the hotter it is, the faster it will use up its fuel.

A white dwarf is supported by the exclusion principle repulsion between the electrons in its matter.

After gravitational collapse a black hole must settle down into a state in which it could be rotating, but not pulsating. Moreover, its size and shape would depend only on its mass and rate of rotation, and not on the nature of the body that had collapsed to form it. This result became known by the maxim: "A black hole has no hair".

If the rays of light that form the event horizon, the boundary of the black hole, can never approach each other, the area of the event horizon might stay the same or increase with time but it could never decrease because that would mean that at least some of the rays of light in the boundary would have to be approaching each other.

Black holes are not really black after all: they glow like a hot body, and the smaller they are, the more they glow. So, paradoxically, smaller black holes might actually turn out to be easier to detect than large ones!

Jacob Bekenstein suggested that the area of the event horizon was a measure of the entropy of the black hole. As matter carrying entropy fell into a black hole, the area of its event horizon would go up, so

that the sum of the entropy of matter outside black holes and the area of the horizons would never go down.

A space-time in which events have imaginary values of the time coordinate is said to be Euclidean.

A scientific theory is just a mathematical model we make to describe our observations: it exists only in our minds. So it is meaningless to ask: which is real, 'real' or 'imaginary' time? It is simply a matter of which is the more useful description.

But if the universe is really completely self-contained, having no boundary or edge, it would have neither beginning nor end: it would simply be. What place, then, for a creator?

Our subjective sense of the direction of time, the psychological arrow of time, is therefore determined within our brain by the thermodynamic arrow of time. Just like a com-puter, we must remember things in the order in which entropy increases.

Quantum mechanics deals with this situation via a class of quantum theories in which particles don't have well-defined positions and velocities but are represented by a wave. These quantum theories are deterministic in the sense that they give laws for the evolution of the wave with time.

Thus if one knows the wave at one time, one can calculate it at any other time. The unpredictable, random element comes in only when we try to interpret the wave in terms of the positions and velocities of particles. But maybe that is our mistake:

maybe there are no particle positions and velocities, but only waves. It is just that we try to fit the waves to our preconceived ideas of positions and velocities.